

2.1 Equations of Motion

Question Paper

Course	CIEA Level Physics
Section	2. Kinematics
Topic	2.1 Equations of Motion
Difficulty	Medium

Time allowed: 20

Score: /10

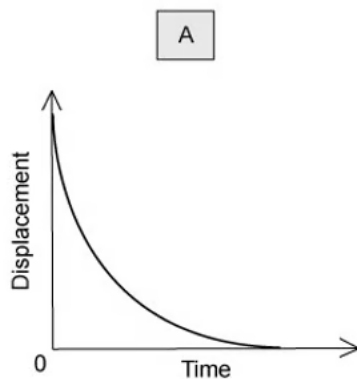
Percentage: /100

Question 1

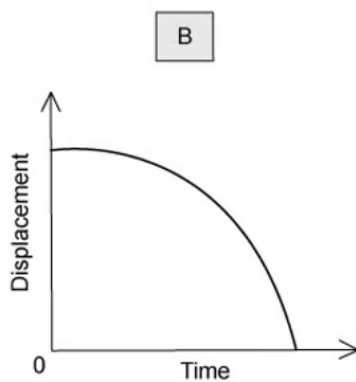
A sphere is released and falls. Its initial acceleration reduces until it eventually begins to travel at constant terminal velocity.

Which displacement-time graph best represents the motion of the sphere?

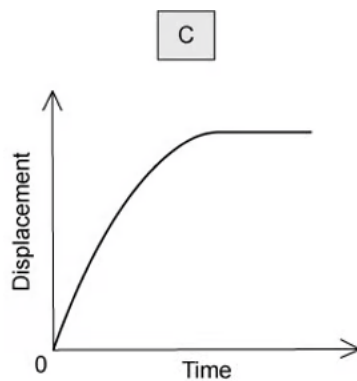
A.



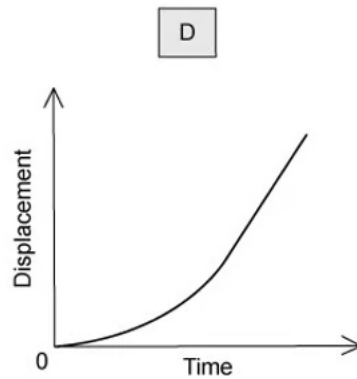
B.



C.



D.



[1 mark]

Question 2

A bullet is fired horizontally with speed v from a rifle. For a short time t after leaving the rifle, the only force affecting its motion is gravity. The acceleration of free fall is g .

Which expression gives the value of $\frac{\text{the horizon distance travelled in time } t}{\text{the vertical distance travelled in time } t}$?

A. $\frac{vt}{g}$

B. $\frac{v}{gt}$

C. $\frac{2vt}{g}$

D. $\frac{2v}{gt}$

[1 mark]

Question 3

The acceleration of free fall on the Moon is one-sixth of that on Earth. On Earth it takes time t for a stone to fall from rest a distance of 2 m.

What is the time taken for a stone to fall from rest a distance of 2 m on the Moon?

A. $6t$

B. $\frac{t}{6}$

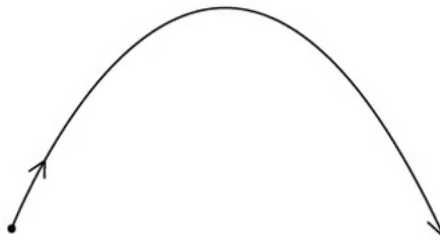
C. $t\sqrt{6}$

D. $\frac{t}{\sqrt{6}}$

[1 mark]

Question 4

A stone is thrown upwards and follows a curved path.



Air resistance is negligible.

Why does the path have this shape?

A. the stone has a constant horizontal velocity and constant vertical acceleration

B. the stone has a constant horizontal acceleration and constant vertical velocity

C. the stone has a constant upward acceleration followed by a constant downward acceleration

D. the stone has a constant upward velocity followed by a constant downward velocity

[1 mark]

Question 5

A train passes through a station at a constant speed of 20 m s^{-1}

A second train is at rest at the station. It begins to leave the station with a uniform acceleration of 1.0 m s^{-2} just as the first train goes past.

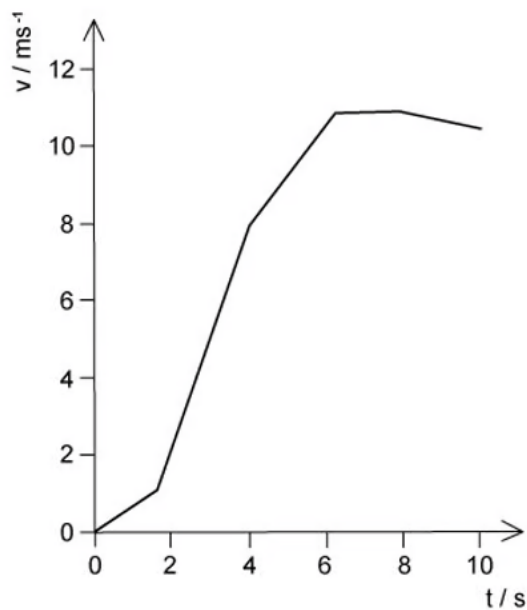
How much time passes before the second train overtakes the first train?

- A. 5 s
- B. 10 s
- C. 20 s
- D. 40 s

[1 mark]

Question 6

The graph shows how the speed v of a sprinter changes with time t during a 100 m race.



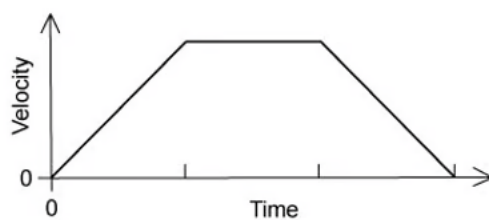
What is the best estimate of the maximum acceleration of the sprinter?

- A. 3 ms^{-2}
- B. 6 ms^{-2}
- C. 9 ms^{-2}
- D. 12 ms^{-2}

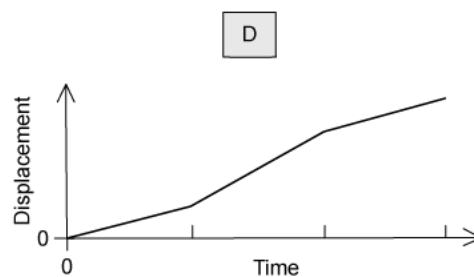
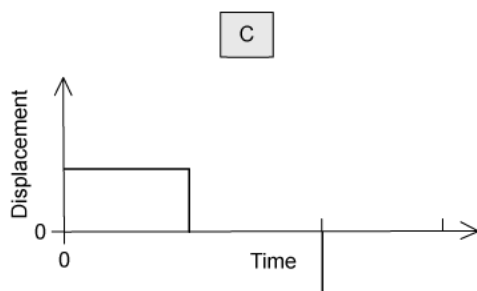
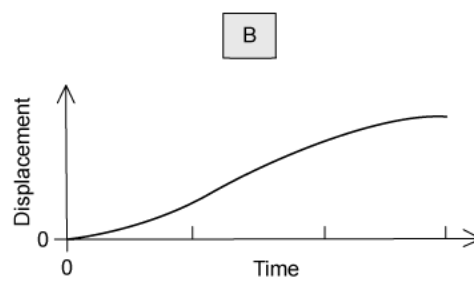
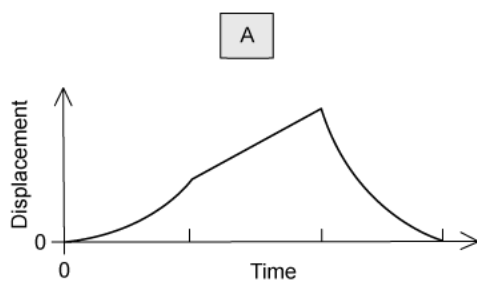
[1 mark]

Question 7

The graph of velocity against time for an object moving in a straight line is shown



What is the corresponding graph of displacement against time?



[1 mark]

Question 8

In a cathode-ray tube, an electron is accelerated uniformly in a straight line from a speed of $4 \times 10^5 \text{ m s}^{-1}$ to one tenth of the speed of light over a distance of 10 mm

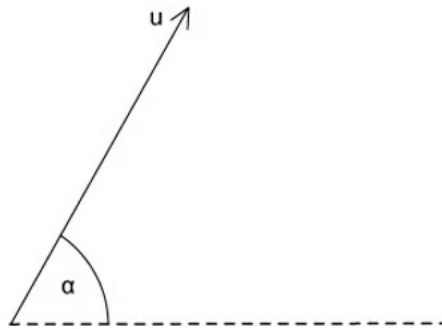
What is the acceleration of the electron?

- A. $4.5 \times 10^3 \text{ m s}^{-2}$
- B. $4.5 \times 10^6 \text{ m s}^{-2}$
- C. $4.5 \times 10^{13} \text{ m s}^{-2}$
- D. $4.5 \times 10^{16} \text{ m s}^{-2}$

[1 mark]

Question 9

A projectile is fired at an angle α to the horizontal at a speed u , as shown.



What are the vertical and horizontal components of its velocity after a time t ?

Assume that air resistance is negligible. The acceleration of free fall is g .

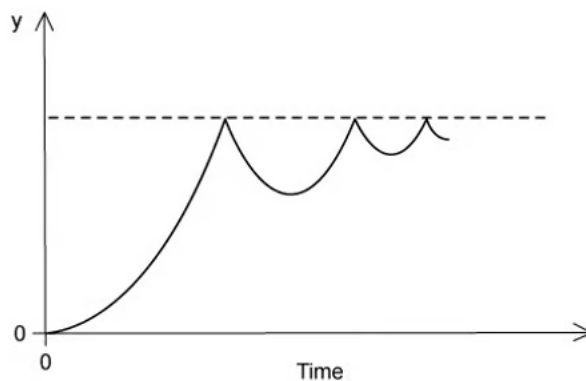
	Vertical component	Horizontal component
A	$u \sin \alpha$	$u \cos \alpha$
B	$u \sin \alpha - gt$	$u \cos \alpha - gt$
C	$u \sin \alpha - gt$	$u \cos \alpha$
D	$u \cos \alpha$	$u \sin \alpha - gt$

[1 mark]

Question 10

A ball is released from rest above a horizontal surface and bounces several times.

The graph shows how, for this ball, a quantity y varies with time.



What is the quantity y ?

- A. acceleration
- B. displacement
- C. kinetic energy
- D. velocity

[1 mark]

Model Answers: Medium

1

The correct answer is **D** because:

- As the ball falls further from its starting position, the displacement increases meaning the gradient increases
- Terminal velocity is the maximum velocity reached by the falling sphere, meaning the first section of the graph should curve upwards to show the initial acceleration and the second section should be a straight line to show the constant velocity

A & B are incorrect because the gradients of these graphs are negative

C is incorrect because the horizontal straight line means the velocity is zero and the ball has stopped moving which is not the case

2

The correct answer is **D** because:

- The acceleration of freefall is downwards and has no effect on horizontal motion
- We can find the horizontal distance travelled using
 - Distance = speed $\times$ time or $s_H = vt$ **(1)**
- Since the bullet is fired in a horizontal direction, the vertical component of the velocity is initially zero
- We can find the vertical distance using
 - $s = ut + \frac{1}{2}at^2$
- $u = 0$ and $a = g$, so
 - $s_V = \frac{1}{2}gt^2$ **(2)**
- Combining equations **(1)** and **(2)** leads us to

$$\begin{aligned} \circ \frac{\text{the horizontal distance travelled in time } t}{\text{the vertical distance travelled in time } t} &= \frac{s_H}{s_V} \\ \circ \frac{s_H}{s_V} &= \frac{vt}{\frac{1}{2}gt^2} = \frac{2v}{gt} \end{aligned}$$

3

The correct answer is **C** because:

Acceleration of free fall on Earth = g



Acceleration of free fall on the Moon = $\frac{g}{6}$



The equation that links s , u , t and a is:

$$s = ut + \frac{1}{2}at^2$$

On earth:

$$u = 0 \quad s = 2\text{m} \quad a = g\text{ms}^{-2}$$

$$s = ut + \frac{1}{2}at^2$$

$$2 = 0 + \frac{1}{2}gt^2$$

$$4 = gt^2$$

$$g = \frac{4}{t^2}$$

On the Moon:

$$u = 0 \quad s = 2\text{m} \quad a = \frac{g}{6}\text{ms}^{-2}$$

$$s = ut + \frac{1}{2}at^2$$

$$2 = 0 + \frac{1}{2}\left(\frac{g}{6}\right)t_M^2$$

$$24 = gt_M^2$$

$$t_M = \sqrt{\frac{24}{g}}$$

Combining these two equations gives:

$$t_M = \sqrt{\frac{24}{g}} = \sqrt{6t^2} = t\sqrt{6}$$

4

The correct answer is **A** because:

- The horizontal and vertical components of velocity and acceleration can be treated independently
- In the absence of air resistance, the horizontal component of velocity is constant while the vertical component of velocity downwards increases at a constant acceleration of 9.81 m s^{-2} (due to the force of gravity)

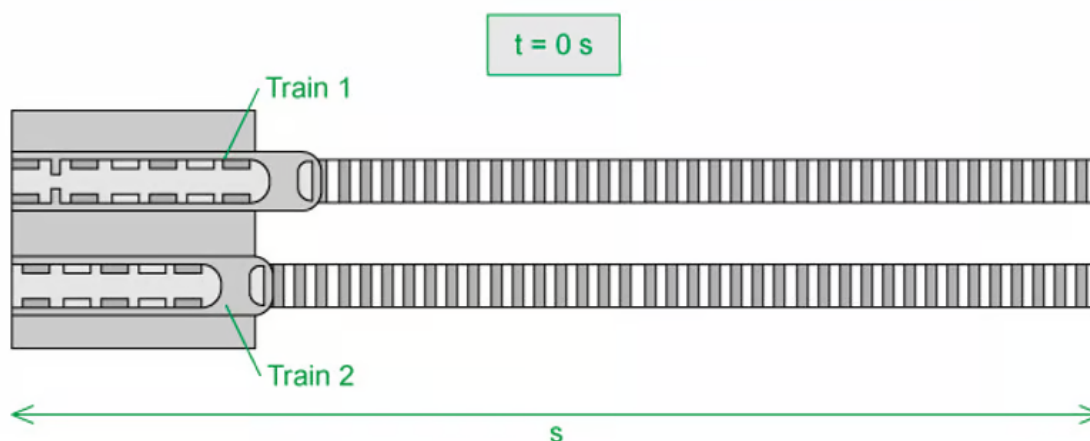
B & D are incorrect because once the stone has been thrown, it is no longer being acted on by the force that propelled it into the air so there is no horizontal acceleration. The force of gravity causes the stone to decelerate upwards and accelerate downwards at a constant rate meaning the vertical velocity is not constant

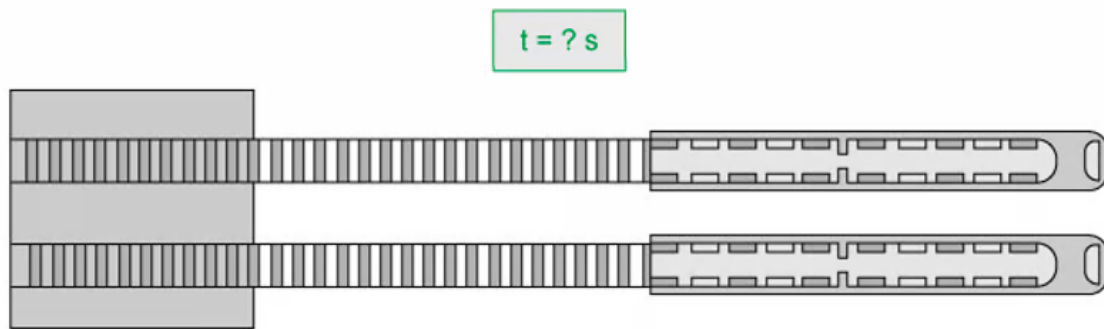
C is incorrect because the stone has a constant upward **deceleration** followed by a constant downward acceleration

5

The correct answer is **D** because:

- Train 2 leaves the station as train 1 passes, so the station can be taken as the origin for the time and distance
- For train 2 to overtake train 1, the distance, s , travelled by both trains must be equal and the time, t , is how long it takes for this to happen





Train 1

Train 1 travels at a constant speed of 20 ms^{-1}

distance = speed $\times$ time

$$s = 20t$$

Train 2

$$a = 1.0 \text{ ms}^{-2} \quad u = 0 \text{ ms}^{-1} \quad s = 20t \quad t = ?$$

The equation that links these quantities is:

$$s = ut + \frac{1}{2}at^2$$

$$20t = 0 + \frac{1}{2}(1)t^2$$

$$40t = t^2$$

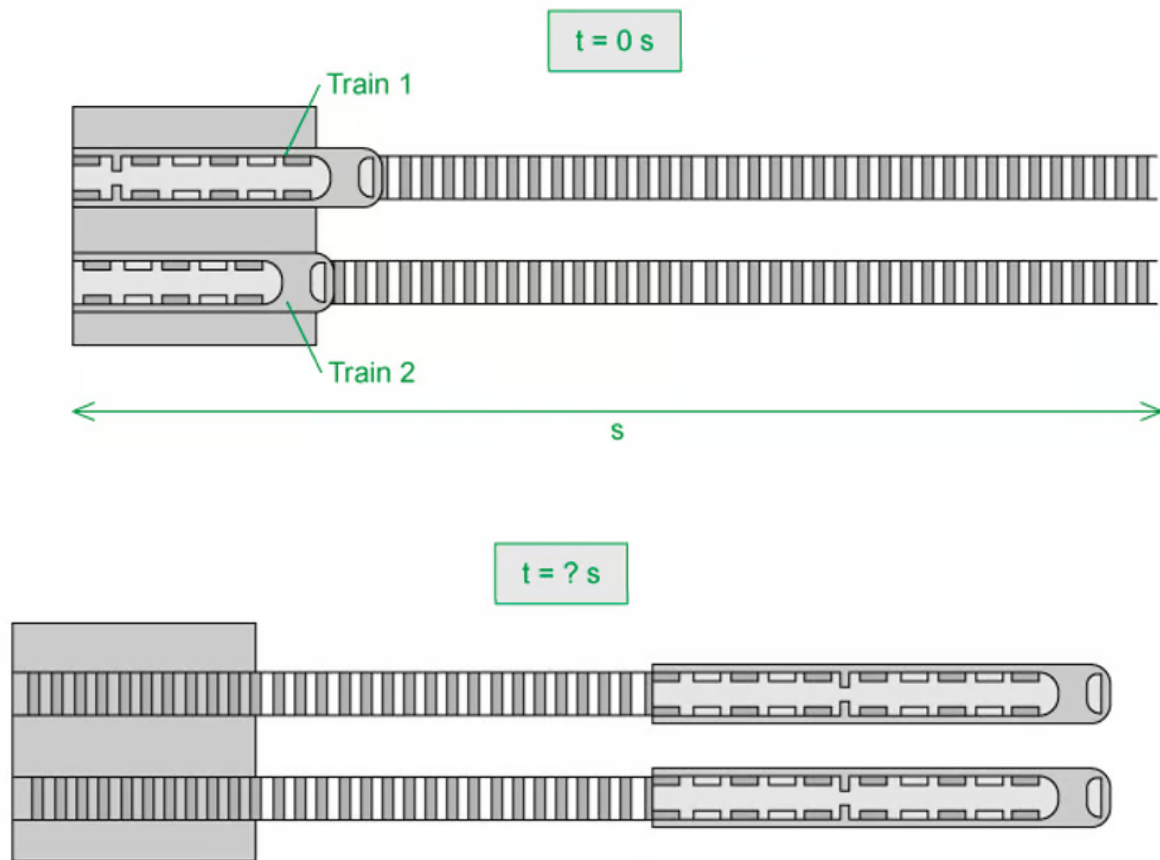
$$t = 40 \text{ s}$$

6

The correct answer is **A** because:

- Acceleration is represented by the gradient of the velocity-time graph
- The steeper the gradient the higher the acceleration, so to find the maximum acceleration we need to estimate the gradient of the steepest line

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Train 2

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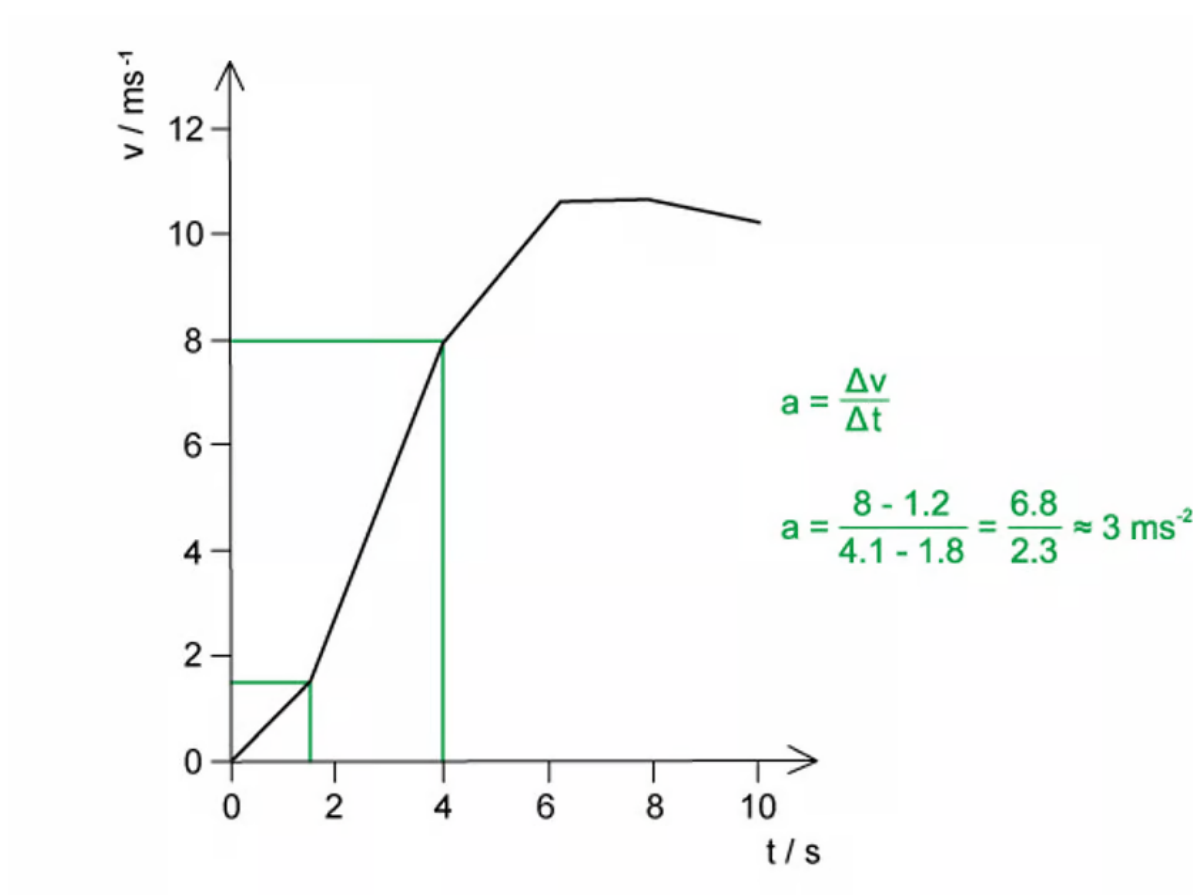
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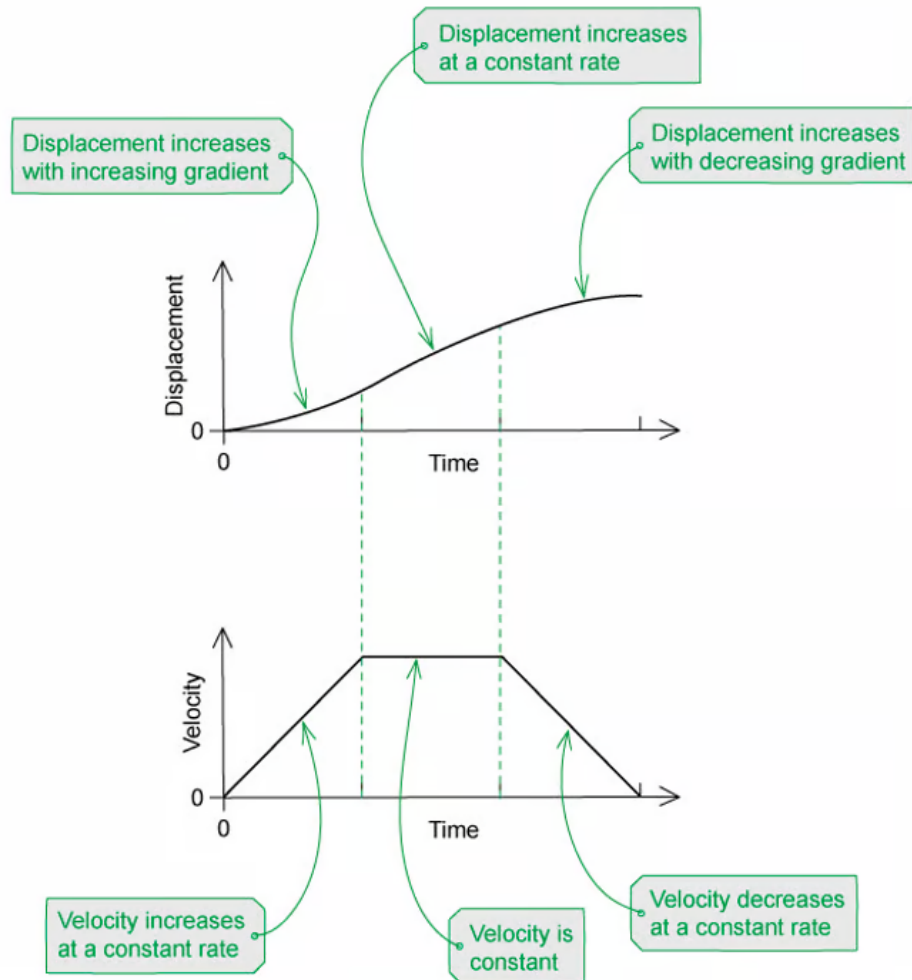
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7

The correct answer is **B** because:

- The gradient of the velocity-time graph gives the acceleration and the area under the graph gives the displacement
- The velocity is always positive, so the displacement is always positive
- The gradient of the displacement-time graph gives the velocity



8

The correct answer is **D** because:

- First let's write out the quantities we have been given
 - $u = 4 \times 10^5 \text{ m s}^{-1}$
 - $s = 10 \text{ mm} = 10 \times 10^{-3} \text{ m}$
 - $v = 3 \times 10^8 \times \frac{1}{10} = 3 \times 10^7 \text{ m s}^{-1}$
 - $a = ?$
- The equation that links u , v , s and a is
 - $v^2 = u^2 + 2as$
- We need to rearrange this to find a
 - $2as = v^2 - u^2$
 - $a = \frac{v^2 - u^2}{2s}$

$$\circ \quad a = \frac{(3 \times 10^7)^2 - (4 \times 10^5)^2}{2 \times 10 \times 10^{-3}} = 4.5 \times 10^{16} \text{ m s}^{-2}$$

9

The correct answer is **C** because:

- We can use trigonometry to deduce
 - Horizontal ($\rightarrow$) component = $u \cos \alpha$
 - Vertical ($\uparrow$) component = $u \sin \alpha$
- Because of gravity, we have to consider the free fall of acceleration, which is equal to g
- This only affects the vertical component of velocity, and acts in the opposite direction to the motion
- So, at a time, t , the vertical component of velocity = $u \sin \alpha - gt$
- The horizontal component is not influenced by gravity so is equal to $u \cos \alpha$

10

The correct answer is **B** because:

- If we take the point when the ball is at rest as the origin and the horizontal surface as its maximum displacement
- The graph shows an increasing and decreasing gradient as the ball accelerates and decelerates with each bounce, which is indicative of a displacement-time graph
- Each time the ball strikes the surface, it loses some energy through sound and heat, which is why its displacement decreases each time but does not come to zero as it does not reach its original height

A is incorrect because the acceleration is constant, so the lines would be straight, not curved.